Data Visualization & Analytics Lab

Experiment No. : 3

Perform data transformation operations in PowerBI

Experiment No.3

1. **Aim:** Perform data transformation operations in PowerBI.

2. **Objectives:**
    - Understand why visualization is an important part of data analysis.
    - Understand the components involved in visualization design.

- **Course Outcomes:** Students will be able to apply visualization tools / techniques for various data preprocessing and filtering tasks.

4. **Hardware / Software Required:** PowerBI
5. **Theory:**

Data transformation is the process of converting, cleaning, and structuring raw data into a format that is suitable for analysis, storage, or integration with other systems, ultimately improving data quality and usability.

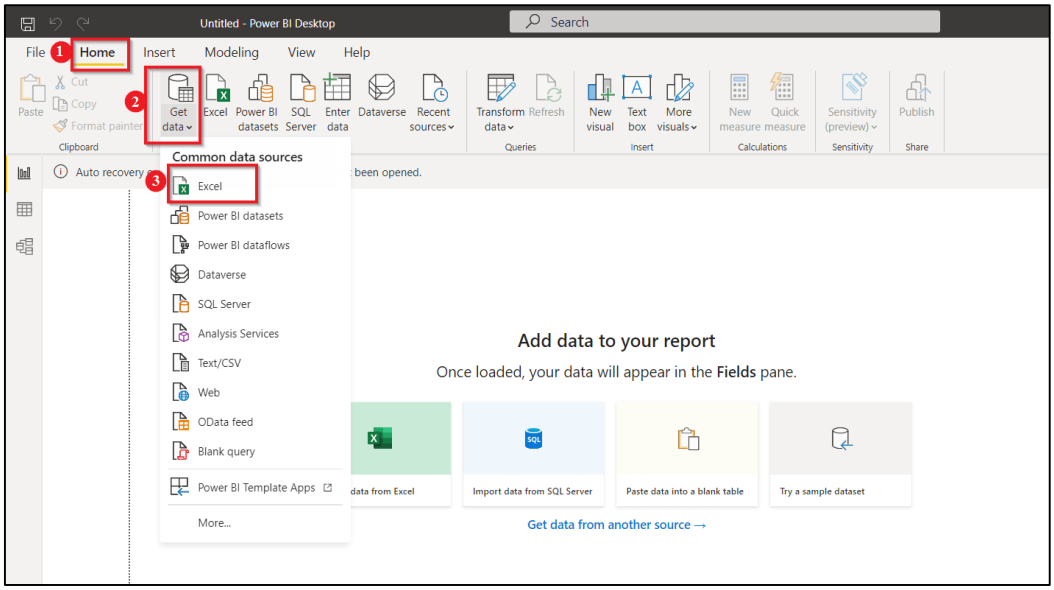
Power BI transforms data in a correlative and interactive manner to analyze, extract and visualize it.

Power Query Editor

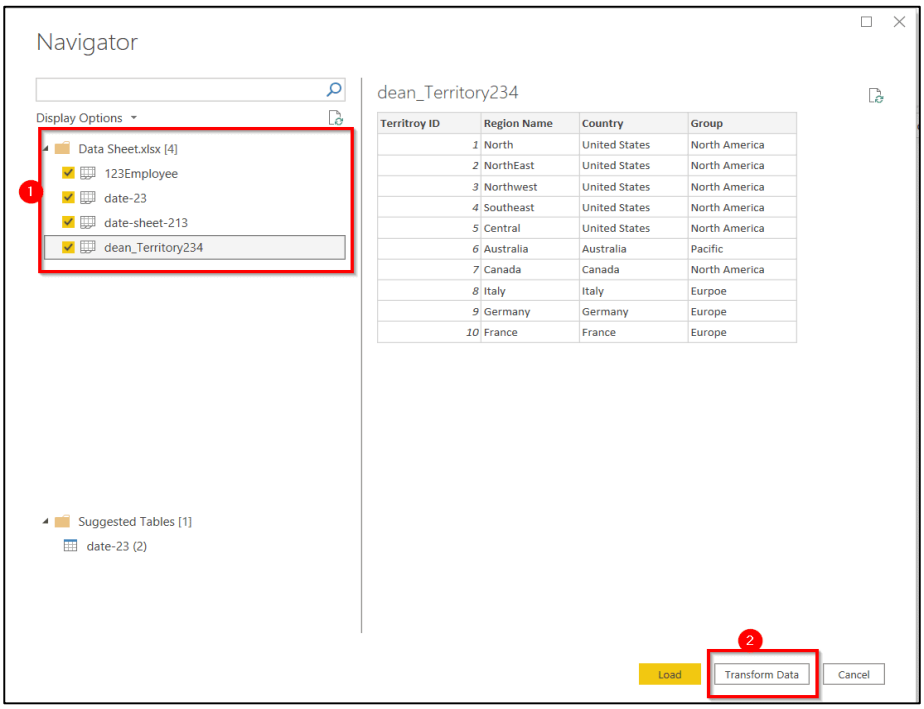
It is the built-in Business Intelligence instrument present in Power BI Desktop to **clean and transform** your data. Any changes in the data are made with the help of Power Query Editor.

Steps:

1. Click on "**Get Data**," it will provide you with the options to source the data from a different platform.



2. Click on "**Transform Data**" option.



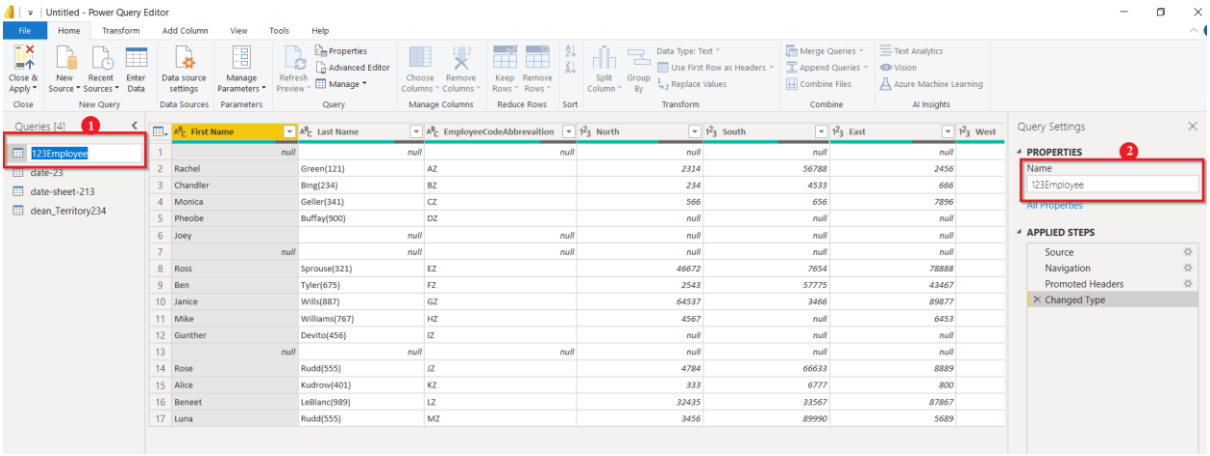
Territroy ID	Region Name	Country	Group
1	North	United States	North America
2	NorthEast	United States	North America
3	Northwest	United States	North America
4	Southeast	United States	North America
5	Central	United States	North America
6	Australia	Australia	Pacific
7	Canada	Canada	North America
8	Italy	Italy	Eurpoe
9	Germany	Germany	Europe
10	France	France	Europe

It will open a **Power Query Editor,** where we will clean and transform our data.

3. Perform the following data transformation steps:

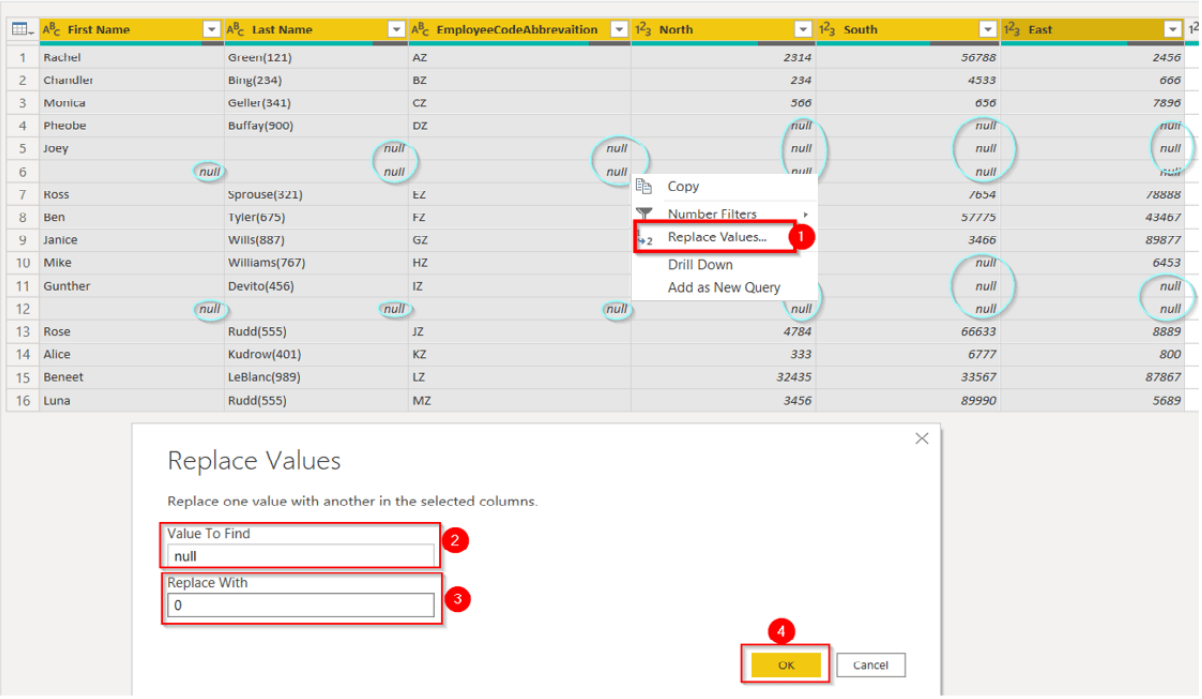
Changing the Table Name

Just double-click the name you want to change and specify the new name, and there you go.



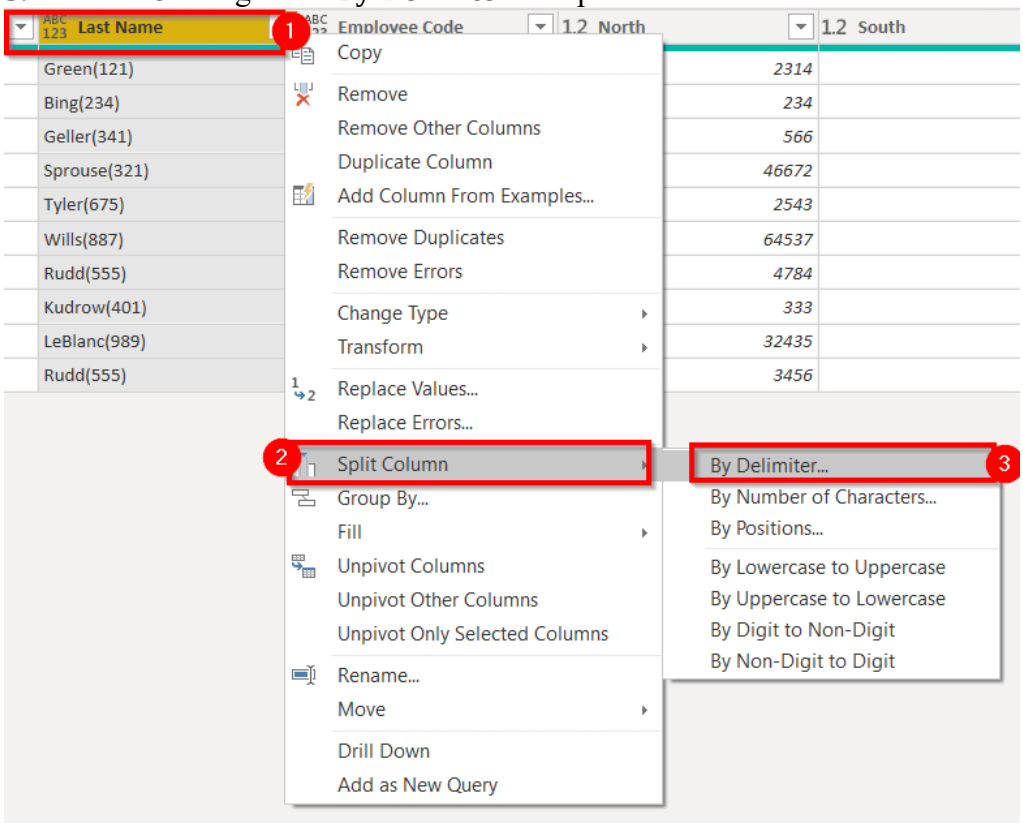
Replace Null Value

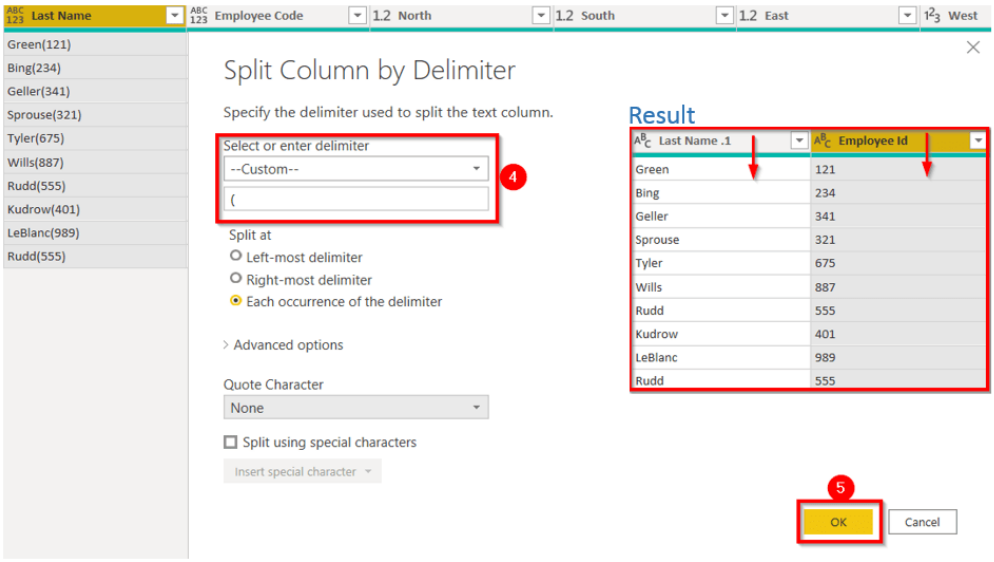
Select all the columns you want to replace the value, "**Right-Click,**" on any row containing the null value. Select "**Replace Values.**"



Separating into two Columns

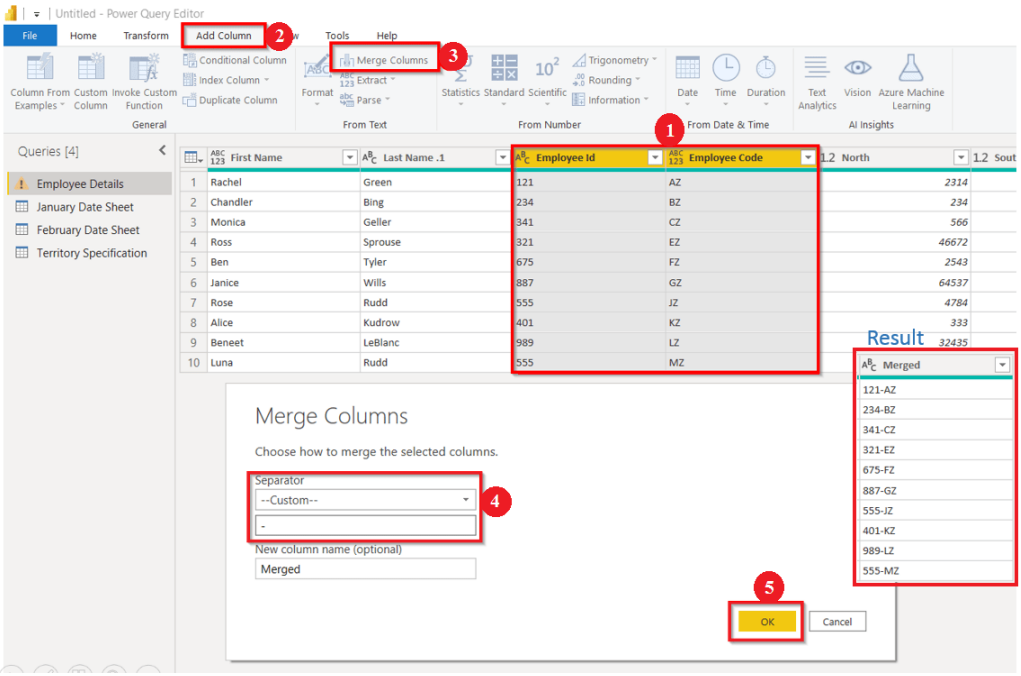
1. Select the column "**Right-Click**" on the column heading.
2. A drop-down list will appear; select **"Split Column**."
3. Then we go for "**By Delimiter**" to split the column.





Merge Columns

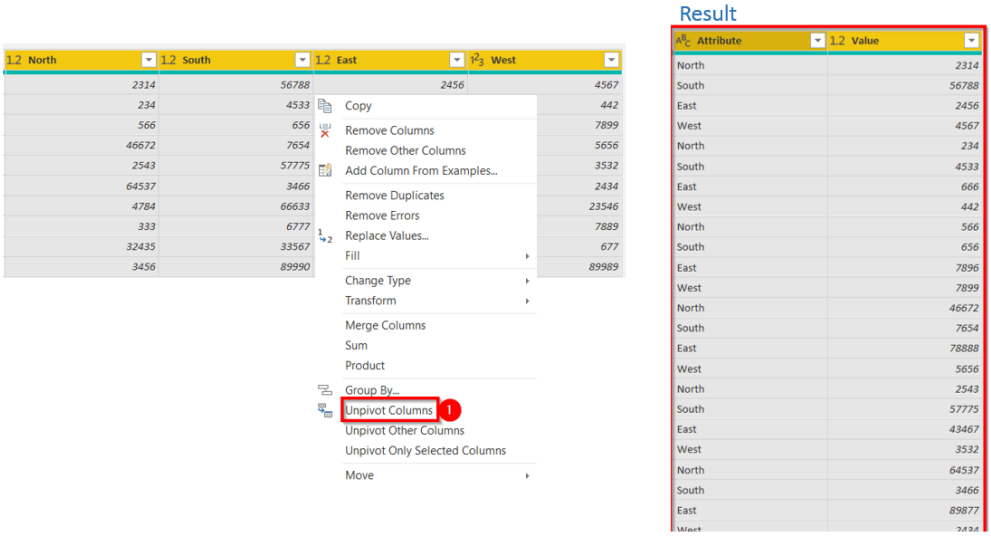
1. Select the two columns you want to merge.
2. Go to "**Add Column.**"
3. Select the "**Merge Columns**" option.
4. Merge Columns page will appear; specify the separator.
5. Click on "**OK.**"



Unpivot the Columns

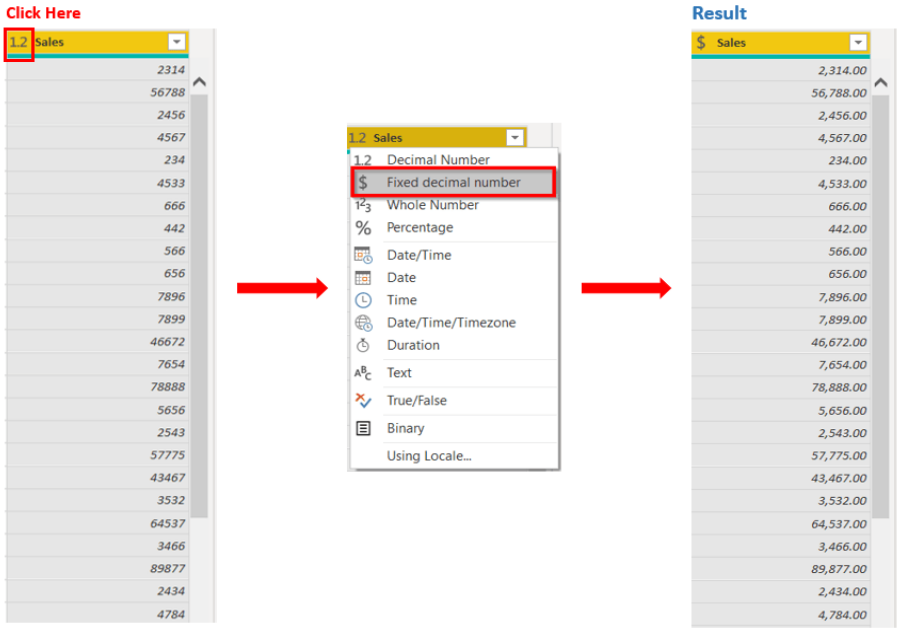
Unpivot column means to unpack the similar values and group them under a single heading. E.g. the four columns North, South, East, and West, show the same metric. Therefore we want to separate it into two columns where one column will show Region and the other Sale.

Select all four columns, **Right-click** on any heading, and then select "**Unpivot Columns.**"



Change Data Type

1. Select the column and "**Right-Click**" on the column header.
2. A drop-down list will appear; select "**Change Type**."
3. Now, again a drop-down list will appear with different data types. Select the data type according to you.



7. Conclusion: Power Query Editor is a potent tool to clean and transform your data. data transformation is essential for effective data visualization as it ensures data is clean, accurate, and in a format suitable for analysis and clear representation.

8. Viva Questions:

- What is data transformation? Why it is important?
- What are different data transformation steps?

References:

1. https://learn.microsoft.com/en-us/power-query/power-query-what-is-power-query
2. Jeremey Arnold," Learning Microsoft Power Bi: Transforming Data Into Insights", O'Reilly